Exercises: Tree-based Classification

Exercise 1 (Feature selection): Consider the following training examples for a binary classification problem, where we want to predict the value of Class attribute (Co or C1?)

Table 4.1. Data set for Exercise 2.

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

(a) Consider Gender, CarType, and ShirtSize, which attribute is the best for predicting Class(C0/C1)? Justify your answer in terms of Information-Gain.

(b) Explain why Customer ID should not be used as the attribute test condition even though it has the lowest Entropy.

Exercise 2 (Building decision tree): Given the following training data: *Buy Computer* data

ID	age	income	student	credit_rating	buy
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31 . . . 40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31 . . . 40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31 . . . 40	medium	no	excellent	yes
13	31 . . . 40	high	yes	fair	yes
14	>40	medium	no	excellent	no

a) Step by step, build a decision tree using Information-Gain as splitting criteria
b) Use the constructed tree to predict the class of the following new example: age<=30, income=medium, student=yes, credit-rating=fair.